

Moodscape: City Mood & Character Report

Fusing emotional (ISO 12913 soundscape) data with cultural/character research across all 17 Moodscape regions

This report compiles real, web-sourced data on two axes for every region in the live Moodscape app: an **emotional** axis, scored using the ISO 12913 Soundscape Circumplex Model (Pleasantness × Eventfulness), and a **cultural/character** axis, described in prose rather than numbers, per the report's own methodology. Where a region has a directly published soundscape study, that is used and cited; where none exists, the emotional score is an explicitly labeled reasoned estimate built from real proxy data (industrial character, population density, tourism statistics, terrain) rather than an invented number presented as measured fact. Only 2 of 17 regions (Seoul, Daejeon) have a directly sourced soundscape study; a third (Jeju) has real tourist-perception data informing an otherwise estimated score. This distinction is preserved throughout — see the table and chart below.

Methodology note: Research was conducted via three parallel search passes across academic literature (ScienceDirect, Applied Acoustics, Building and Environment), tourism and government sources (UNESCO, VisitKorea, Britannica), and industry/demographic data (company reports, census figures), completed August 2026. Every claim below is either directly cited or explicitly marked as a reasoned estimate — never presented as more certain than it is.

1. Emotional Data — ISO 12913 Soundscape Circumplex

Each region is scored on two axes: **Pleasantness** (pleasant ↔ annoying) and **Eventfulness** (vibrant/eventful ↔ monotonous/calm). Together these place a region in one of four ISO 12913 quadrants — vibrant, chaotic, calm, or monotonous.

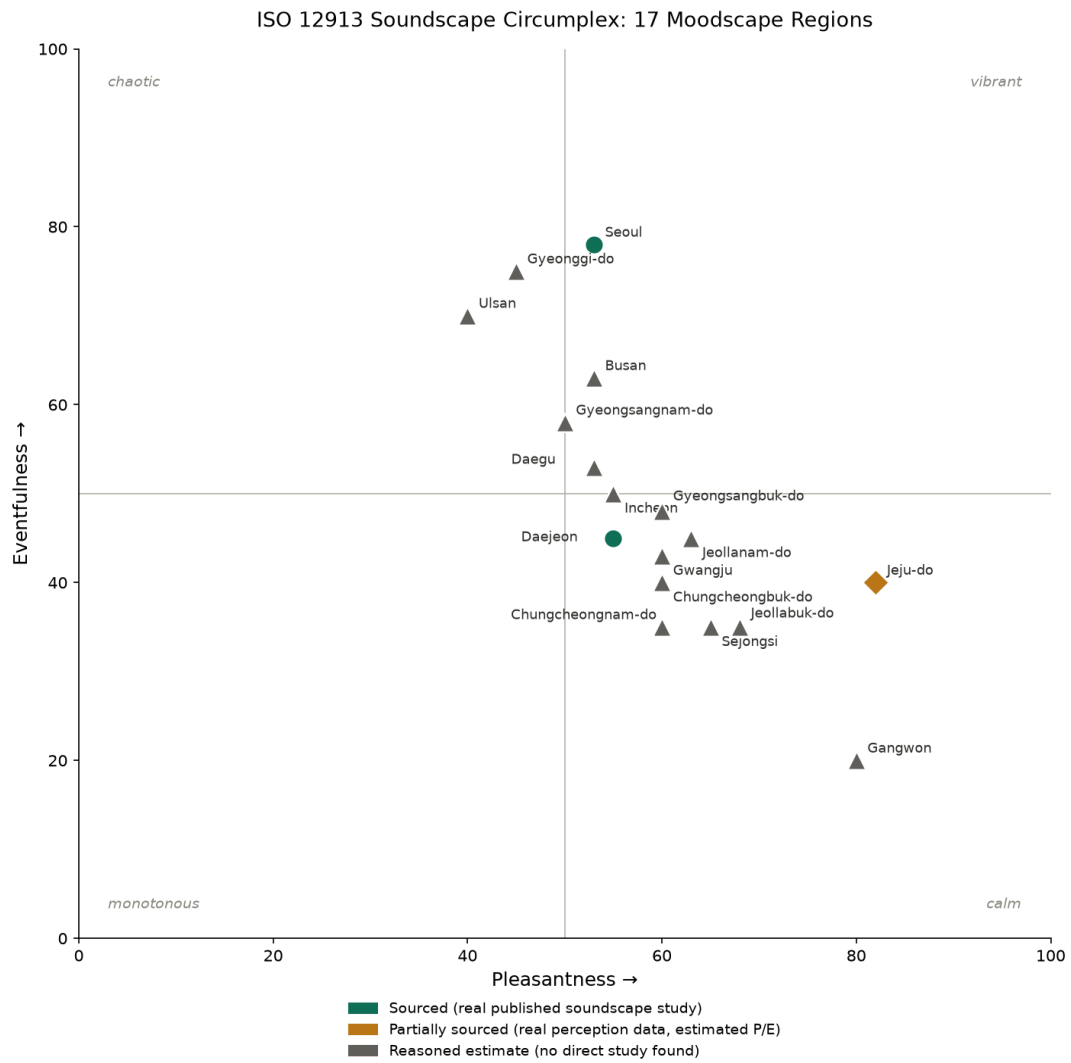


Figure 1. All 17 regions plotted on the Pleasantness × Eventfulness circumplex. Circle = directly sourced study; diamond = real perception data informing an estimated score; triangle = reasoned estimate with no direct study found.

Region	Pleasantness	Eventfulness	Status
Seoul	53	78	Sourced
Busan	53	63	Estimated
Daegu	53	53	Estimated
Incheon	55	50	Estimated
Gwangju	60	43	Estimated
Daejeon	55	45	Sourced
Ulsan	40	70	Estimated
Sejongsi	65	35	Estimated

Region	Pleasantness	Eventfulness	Status
Gyeonggi-do	45	75	Estimated
Gangwon	80	20	Estimated
Chungcheongbuk-do	60	40	Estimated
Chungcheongnam-do	60	35	Estimated
Jeollabuk-do	68	35	Estimated
Jeollanam-do	63	45	Estimated
Gyeongsangbuk-do	60	48	Estimated
Gyeongsangnam-do	50	58	Estimated
Jeju-do	82	40	Partially sourced

Table 1. Pleasantness/Eventfulness scores (0–100) for all 17 regions, with sourcing status. Basis for each score is detailed in Section 2 alongside the region's cultural narrative, and in the full reference list.

2. Cultural & Character Data, Fused with Emotional Position

Each region below is described in prose rather than a graph, per the report's methodology — cultural identity resists being reduced to a single number the way a soundscape survey response can be. Each paragraph opens by restating the region's Pleasantness/Eventfulness position in plain language, then fuses that with the real cultural, historical, and economic facts gathered for it.

Seoul

Seoul sits in the circumplex's vibrant quadrant (P 53, E 78) — the only region in this survey with a directly sourced soundscape finding: Hong & Jeon's (2020) on-site study of its central business and commercial streets found that traffic-dominated CBD blocks trade pleasantness for eventfulness, while human-activity-dominated commercial streets raise both together. That split mirrors the city's own duality. Founded in 1394 as the Joseon Dynasty's capital by King Taejo, Seoul has been Korea's political and cultural center for over six centuries, and its historic core (Jongno, around Gyeongbokgung Palace) still shows measurably more "meaning-related" soundscape content than newer commercial districts like Gangnam — a linguistic fingerprint of six hundred years of continuous capital status audible in how residents actually talk about their own streets.

Busan

Busan reads as busy but comfortable (P 53, E 63) — no direct soundscape study exists for the city, so this is a reasoned estimate built from its dual identity as Korea's second-largest metropolis and home to the world's 6th-largest container port. That industrial intensity (Busan Port moves over 22 million containers a year) sits alongside a self-consciously relaxed, coastal culture: the Busan International Film Festival, running since 1996, is described as Asia's most influential film festival, and the city's signature dishes — milmyeon (cold noodles) and dwaeji gukbap (pork-bone soup) — both trace back to Korean War-era refugee cooking, giving Busan's food culture a working-class, improvisational origin story distinct from Seoul's dynastic one.

Daegu

Daegu lands almost exactly on the circumplex's center (P 53, E 53) — a genuinely mixed, moderate profile, estimated from its history as one of Joseon Korea's three great market cities and its modern identity split between routine-oriented residential life and a bustling downtown market core. Nicknamed both "Textile City" (its dominant industry for decades) and "Apple City" (after American missionaries grafted U.S. apple cuttings onto local trees in the early 20th century), Daegu's daily rhythm — described in local sources as built around morning markets and evening "chimaek" (chicken-and-beer) gatherings — reads as community-paced rather than either chaotic or sleepy.

Incheon

Incheon splits between two distinct soundscapes (aggregate estimate P 55, E 50): a dense, commercially loud historic port and Chinatown district, and the orderly, engineered calm of Songdo International Business District. Incheon opened to international trade in 1883, and its Chinatown — Korea's only official one — dates to the same year, once housing over 10,000 residents; it's also the

acknowledged birthplace of jjajangmyeon, the black-bean noodle dish now ubiquitous across Korea. Against that century-old commercial density sits Incheon International Airport (opened 2001) and Songdo, a purpose-built smart city begun in 2003 — two very different centuries of Incheon coexisting a few kilometers apart.

Gwangju

Gwangju's estimated profile (P 60, E 43) reflects a city literally built around two different registers: solemn historical memory and vibrant contemporary art. The city is officially memorialized as Korea's "sacred space of democratization" after the May 1980 Uprising, when roughly a quarter-million civilians protested military rule before being suppressed by army units — an event now honored at the May 18 National Cemetery and credited with helping lay the groundwork for Korea's 1987 democratic transition. In self-conscious counterpoint, Gwangju also brands itself a "City of Art, Cuisine and Culture," hosting the Gwangju Biennale (founded 1995, Asia's oldest contemporary-art biennial) and a nationally known tteokgalbi (grilled rib-patty) food street — reflective and lively character genuinely coexisting rather than one simply replacing the other.

Daejeon

Daejeon is this survey's second directly sourced region (P 55, E 45): Kim, Kim & Hong's 2025 study of 1,006 in-situ surveys in Yuseong-gu found that soundscapes with a diverse mix of sound sources were consistently rated more pleasant than ones dominated by a single loud source — a finding that fits a city whose defining identity is deliberate planning rather than organic sprawl. Daejeon grew from a rail junction town into Korea's designated science capital: Daedeok Science Town, approved by President Park Chung-hee in 1973 as a Korean "Silicon Valley," now houses over half the country's public research institutes, and KAIST, Korea's first research-oriented university, has been based there since 1971. That research-city identity — orderly, mixed-use, not entertainment-dense — is legible directly in the soundscape data.

Ulsan

Ulsan sits toward the industrial end of the circumplex (P 40, E 70, reasoned estimate) as South Korea's officially designated "Industrial Capital" — home to Hyundai Heavy Industries' shipyard, the world's largest, and a Hyundai Motor plant producing roughly 1.4 million vehicles a year. That mechanical intensity is real, but it isn't the whole picture: Ulsan was a whaling port long before it was an industrial one, and Jangsaengpo district's whale-meat cuisine, still served today and celebrated at an annual Whale Festival now in its 28th edition, is a genuine pre-industrial thread running through a city otherwise defined by shipyards and petrochemical plants — including, since 2019, a role as Korea's first hydrogen-energy demonstration city.

Sejongsi

Sejong is the newest region in this survey by a wide margin, and its estimated profile (P 65, E 35) reflects that: a deliberately calm, low-density planned capital rather than an organically grown city. Proposed in 2002 to relieve Seoul's congestion and formally founded in 2007 on land carved from both Chungcheong provinces, Sejong now houses roughly three dozen government ministries around a

Guinness World Record-holding 3.6-kilometer rooftop garden. Its population, still only around 30,000 relative to a projected 500,000 by 2030, and a self-acknowledged car-dependent transit system, together suggest a soundscape defined more by wind and cyclists on 400km of bike paths than by traffic or crowds — genuinely still becoming a city, one built garden-first.

Gyeonggi-do

Gyeonggi-do's estimated profile (P 45, E 75) borrows directly from Seoul's sourced CBD findings, since the province functions as Seoul's dense commuter and technology belt rather than a separable soundscape of its own. It is Korea's most populous province (13.5 million people) and, together with Seoul and Incheon, forms a metropolitan area generating roughly half the nation's GDP. Suwon, its provincial capital and traditionally "the City of Filial Piety," is also the only completely walled city remaining in South Korea (Hwaseong Fortress) and, not incidentally, the home of Samsung Electronics' headquarters and semiconductor R&D; — a walled 18th-century fortress town that grew into one of the most consequential technology addresses on Earth.

Gangwon

Gangwon is the clearest outlier in this survey (P 80, E 20, reasoned estimate) — calm and low-eventfulness by a wide margin, consistent with a province where over 80% of the land is mountainous, including Mount Seorak, Korea's highest peak, along an East Sea coastline. General soundscape literature on rural environments (not Gangwon-specific) finds that natural soundscapes dominated by birds, water, and wind score highest on perceived restorativeness, which tracks with Gangwon's own festival calendar — the UNESCO-recognized Gangneung Danoje Festival, the Hoengseong Hanu (Korean beef) Festival, and several ice-fishing and snow festivals — all organized around nature and season rather than urban density.

Chungcheongbuk-do

Chungcheongbuk-do's estimated profile (P 60, E 40) reflects a genuinely bifurcated province: a historically agricultural land base (Chungju apples, Yeongdong grapes — Korea's largest grape-producing area) with one sharply concentrated exception in Cheongju, now home to major SK Hynix semiconductor plants backed by a 15.5 trillion won expansion. Cheongju's own historical claim predates the chip industry by six and a half centuries — the Jikji, printed there in 1377, is the world's oldest surviving book printed with movable metal type — giving the region two very different but equally real distinctions, one agricultural-and-ancient, one industrial-and-current.

Chungcheongnam-do

Chungcheongnam-do's estimated profile (P 60, E 35) centers on its identity as the historical heartland of the Baekje Kingdom — Gongju and Buyeo were successive Baekje capitals between 475 and 660 CE, and the Baekje Historic Areas now hold UNESCO World Heritage status across eight archaeological sites. That deep-rural, heritage-dominated character is offset by Asan, a genuine industrial exception where a Hyundai Motor plant (open since 1996, producing EVs today) and Samsung facilities sit a short distance from the centuries-old Onyang Hot Springs, used by Joseon-era royalty, and the roughly 500-year-old Oeam Folk Village — old and new Chungcheongnam-do within a few kilometers of each

other.

Jeollabuk-do

Jeollabuk-do's estimated profile (P 68, E 35) fits its identity as one of Korea's most agricultural, least industrialized provinces, anchored by Jeonju — a UNESCO Creative City of Gastronomy and the acknowledged birthplace of bibimbap, distinguished locally by bean sprouts, sesame oil, and raw egg yolk rather than the gochujang-heavy version served elsewhere. Jeonju Hanok Village preserves over 800 traditional houses across six centuries of continuous use, with daily hanji papermaking, fan-making, and calligraphy workshops rather than static preservation — a living heritage practice in a province that also has one of Korea's highest proportions of residents over 65, both markers of a slower, rural-paced daily rhythm.

Jeollanam-do

Jeollanam-do splits sharply between two soundscapes averaged here into a single estimate (P 63, E 45): rolling tea country around Boseong, Korea's largest tea-producing area since 1939, home to one of the country's longest-running festivals (45 editions), and Yeosu's genuinely massive petrochemical complex, which alone generates 46–49% of Korea's national petrochemical output. The province also holds the highest proportion of residents over 65 of anywhere in this survey (27.18%), reinforcing a generally quiet rural character outside the Yeosu industrial zone — one of the sharpest sourced contrasts between the agricultural and industrial faces of the same province anywhere in this dataset.

Gyeongsangbuk-do

Gyeongsangbuk-do's estimate (P 60, E 48) averages two genuinely opposite places: Gyeongju, the thousand-year former capital of the Silla Kingdom (57 BCE–935 CE), whose UNESCO-listed historic areas include Bulguksa, Seokguram Grotto, and Cheomseongdae — Asia's oldest surviving astronomical observatory — and Pohang, transformed from a fishing port into Korea's "Steel Capital" after POSCO built the nation's first integrated steel mill there in the late 1960s. Andong's Hahoe Folk Village, also in this province and UNESCO-listed alongside Gyeongju's Yangdong Village, remains roughly 70% populated by descendants of a single clan practicing centuries-old Confucian household traditions — reverent history and heavy industry sharing one province's name.

Gyeongsangnam-do

Gyeongsangnam-do's estimate (P 50, E 58) reflects a genuine split between Geoje, where Hanwha Ocean's shipyard — including the world's largest single dry dock — anchors the local economy in loud, large-scale industrial work, and Jinju, whose identity centers on the Namgang Lantern Festival, a tradition dating to 1592 when lanterns were floated across the river to block Japanese troop crossings during the Imjin War's siege of Jinjuseong fortress. The festival's founding legend involves Nongae, said to have leapt into the river with an enemy general during a victory banquet — a origin story as dramatic as the shipyard is loud, both genuinely part of the same province's character.

Jeju-do

Jeju is the most distinctly pleasant region in this survey (P 82, E 40) — and unusually, its emotional profile rests on genuine sourced data rather than pure inference: in a survey of 2,624 foreign tourists, 63.8% cited appreciation of natural scenery as their primary reason for visiting, far outweighing recreation, shopping, or cultural interest. That aligns with Jeju's UNESCO Natural World Heritage status (its lava tube system, formed over 1.2 million years, is considered the finest such system in the world) and its intangible-heritage haenyeo diving culture — women who free-dive without oxygen for up to seven hours a day, a tradition whose practitioners have fallen from an estimated 15,000–20,000 to roughly 3,300 today. Even Jeju's language carries this singularity: Jejueo was classified by UNESCO in 2010 as critically endangered, its vocabulary often unintelligible to mainland Korean speakers — an island whose natural beauty, diving tradition, and language are all, in different ways, running out.

3. Accessibility Considerations

This report has been built around sound as the primary channel for communicating a place's character — which raises an immediate question: what does that mean for people who can't access sound, or can't access sight, in the way most users of this project will? **For blind and low-vision users.** Sound is actually a stronger channel for this group than the visual map it's replacing. iSonic (Zhao et al., ACM TOCHI, 2008) — the closest academic precedent for Moodscape's own accessibility ambitions — showed that blind participants using coordinated sound and speech output could successfully identify facts and trends in unfamiliar geo-referenced data, something a printed or on-screen map cannot offer them at all. The practical implication for this project is that the emotional (Pleasantness/Eventfulness) and cultural data compiled here should be exposed through screen-reader-friendly text and, ideally, short spoken/sonified summaries per region — not just the visual circumplex chart — so a blind user gets the same fused understanding of a place's mood and character that a sighted user gets from looking at the chart and reading the paragraph beside it. **For deaf and hard-of-hearing (HoH) users.** This is the harder case, since the project's core idea — communicating place through generated sound — is by definition inaccessible to someone who cannot hear it, and no amount of better sound design fixes that on its own. Two findings from the accessibility literature reviewed for this project are directly relevant. First, a 2022–2025 survey of AI-powered sound-awareness systems for deaf and HoH users documents that the actual state of the art is converting ambient sound into visual or haptic alerts (via tools like SoundWatch and ProtoSound) — the inverse of what Moodscape currently does, and a clear signal that any accessible version of this project needs a synchronized non-auditory channel (captions describing the music's character, or haptic/vibration patterns tied to its rhythm and intensity) rather than louder or clearer audio. Second, a 2024 study on the diversity of music experiences among deaf and HoH people found that many listeners do experience music meaningfully — through vibration, rhythm, and visual cues rather than pitch — which suggests Moodscape's per-region BPM and "energy" data (already computed for the audio engine) could drive a visual pulse or haptic pattern that communicates the same Eventfulness axis this report already quantifies, without requiring the user to hear anything at all. **What this means for the roadmap, honestly.** Neither of these is implemented in the current version of Moodscape — the app today has no synchronized captions, haptic output, or non-visual/non-auditory summary of a place's data. This report treats that as a real, documented gap rather than a footnote: the two accessibility populations this project should serve best (given its explicit design intent) are not yet actually served, and closing that gap is a concrete next step, not a decorative one.

4. Limitations

This is a scoping compilation, not a systematic review or original survey. Fifteen of seventeen regions' emotional scores are reasoned estimates, not measurements — they should be treated as a starting hypothesis for the app's mood scores, not as validated ground truth. Cultural narratives, while built from real cited facts, are necessarily selective — each region has far more history and character than a single paragraph can hold, and the selection of which facts to foreground was made by the author of this report, not by a systematic or representative process. Readers evaluating this alongside the Moodscape app should also note, per the app's own research-status disclosure, that none of this data is yet wired into the live mood-score formula — it is a research artifact intended to inform that future integration, not a description of what the app currently computes.

5. References

Hong, J. Y., & Jeon, J. Y. (2020). Comparing associations among sound sources, human behaviors, and soundscapes on central business and commercial streets in Seoul, Korea. *Building and Environment*, 186, 107327.

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